

SYSTEM STATUS: ACTIVE

LAT SS RE
LON S/56



LAT SS RE
LON S/56



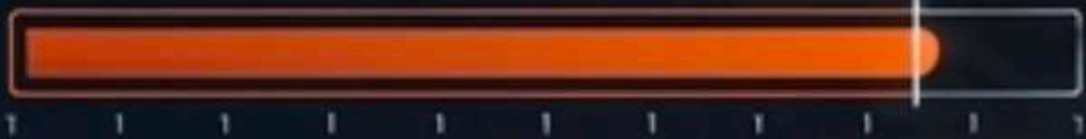
LAT SS RE
LON S/56

Intelligent Solutions for Solar Energy

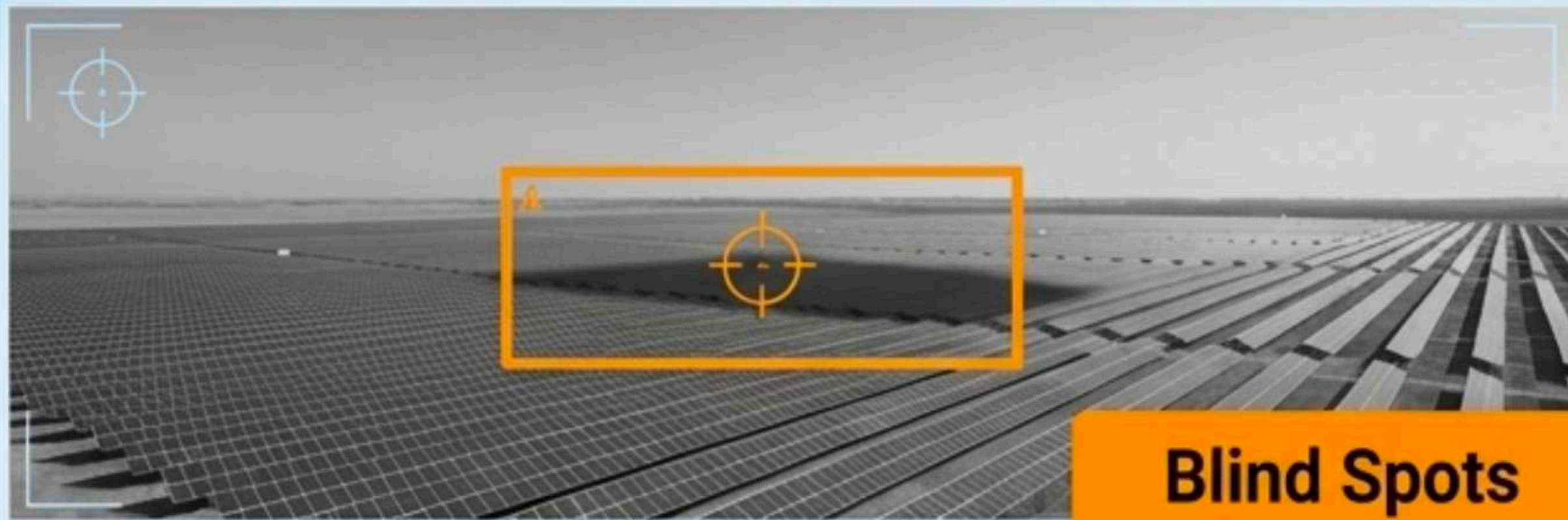
The evolution of O&M powered by
autonomous UAVs and AI

SCANNING...

87%



The scale of solar parks makes manual inspection inefficient



Blind Spots

Vast territories and complex site geography.



Wasted Time

Slow and expensive manual inspections.



Generation Loss

Late detection of hidden defects and anomalies.



Fragmented O&M

Challenges in centralized control and inefficient O&M.

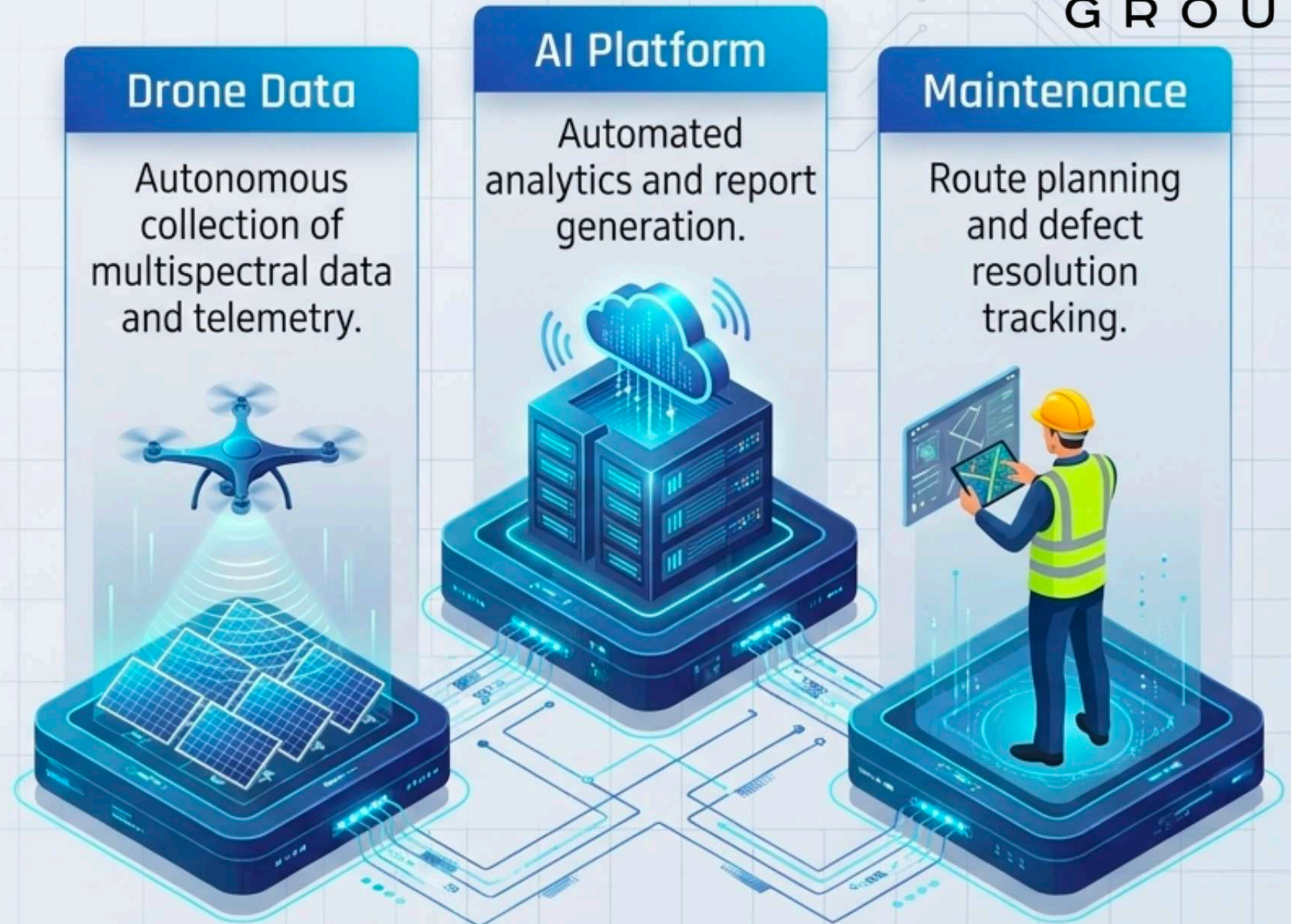
Unified Digital System for O&M

A comprehensive solution covering the entire inspection cycle. This is not just a drone or standalone software — it is a unified operating system for managing solar assets.



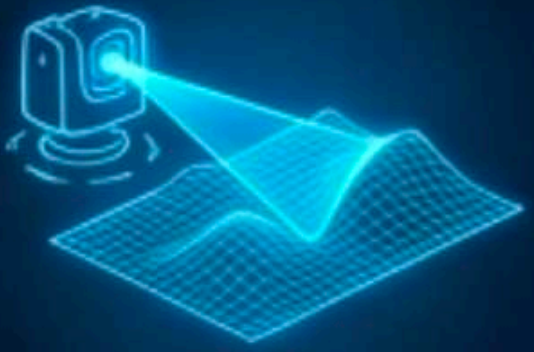
**DRONE
SERVICE**

**UNB
GROUP**



Digital Preparation: Centimeter-Level Precision

Drone data is used for site preparation, building high-precision maps, and environment modeling before the first flight.



LiDAR & Photogrammetry

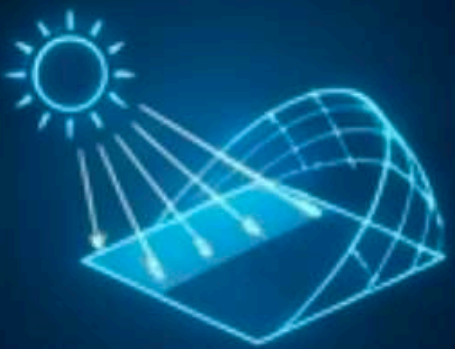
Building a digital twin

WITH CM-LEVEL PRECISION



3D Modeling

Creating an exact replica of the terrain and station assets.



Insolation Analysis

Calculating illumination and shading zones.



Layout Modeling

Optimizing panel placement

TO MAXIMIZE GENERATION



Hardware Level: 100% Autonomy

0 Operators

Takeoff and landing
without constant human
intervention.

Smart Hangar

Automated charging and battery replacement.

24/7 Operation

Weather protection and readiness for scheduled regular inspections.

Safety First

Automatic return upon low battery charge.

Multi-Level Navigation and Flight Safety



**DRONE
SERVICE**

**UNB
GROUP**

01

Automated Planning

Route construction based on 3D models (Point cloud) and camera parameters.

02

Safety Verification

Automated route checking for obstacles and no-fly zones before launch.

03

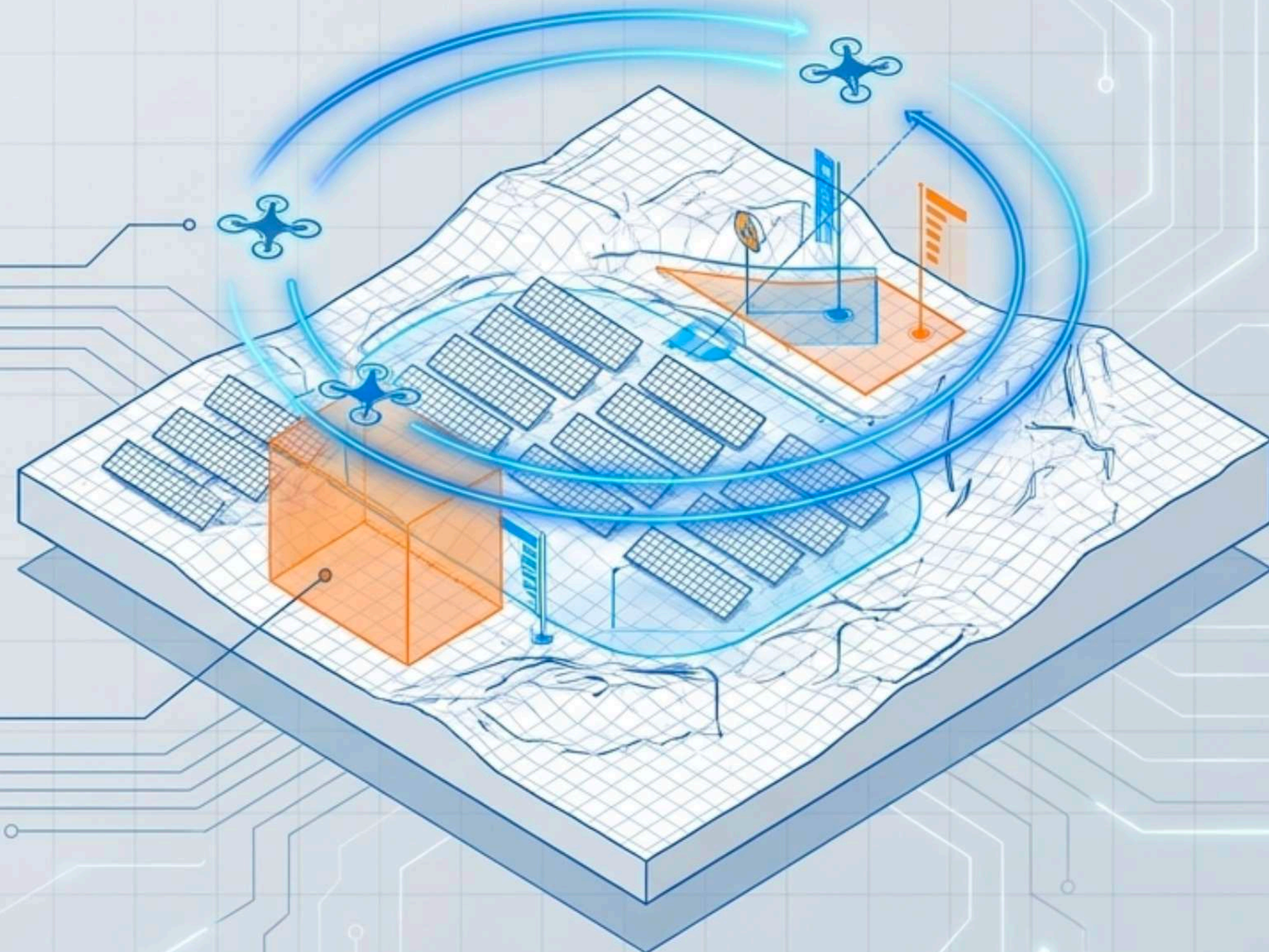
Preview / Simulation

Virtual simulation and mission visualization prior to actual flight.

04

Total Coverage

Adjusting capture points (GSD, overlap, altitude) for 100% site coverage.



System Brain: Multispectral AI Analysis



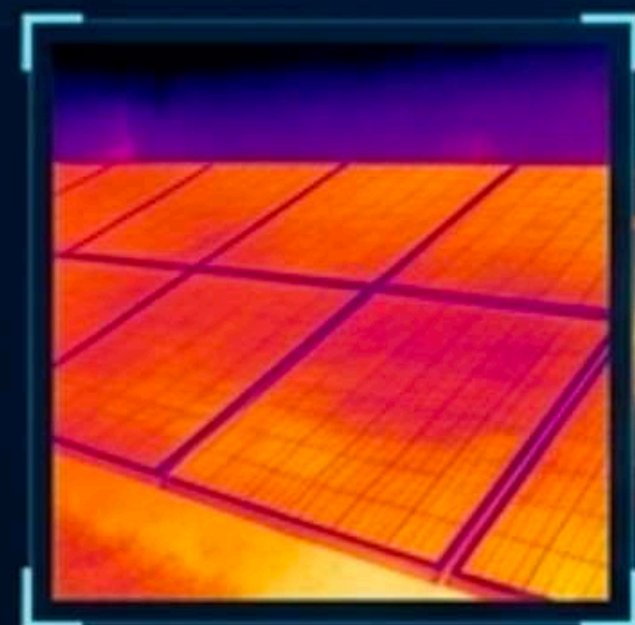
DRONE
SERVICE

UNB
GROUP

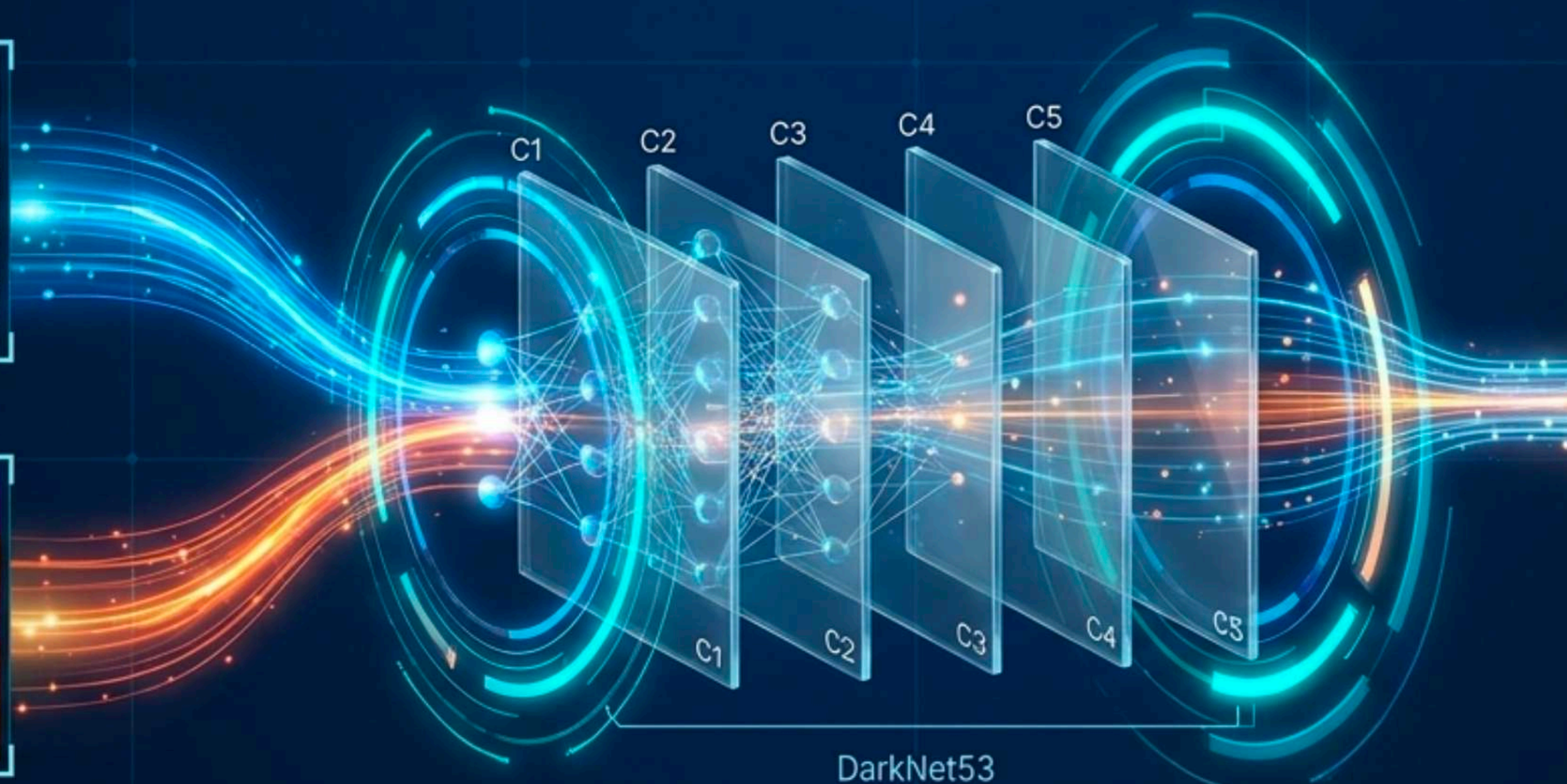
Deep analysis in the visible spectrum (Visible Light) and thermal integration (Image Fusion).



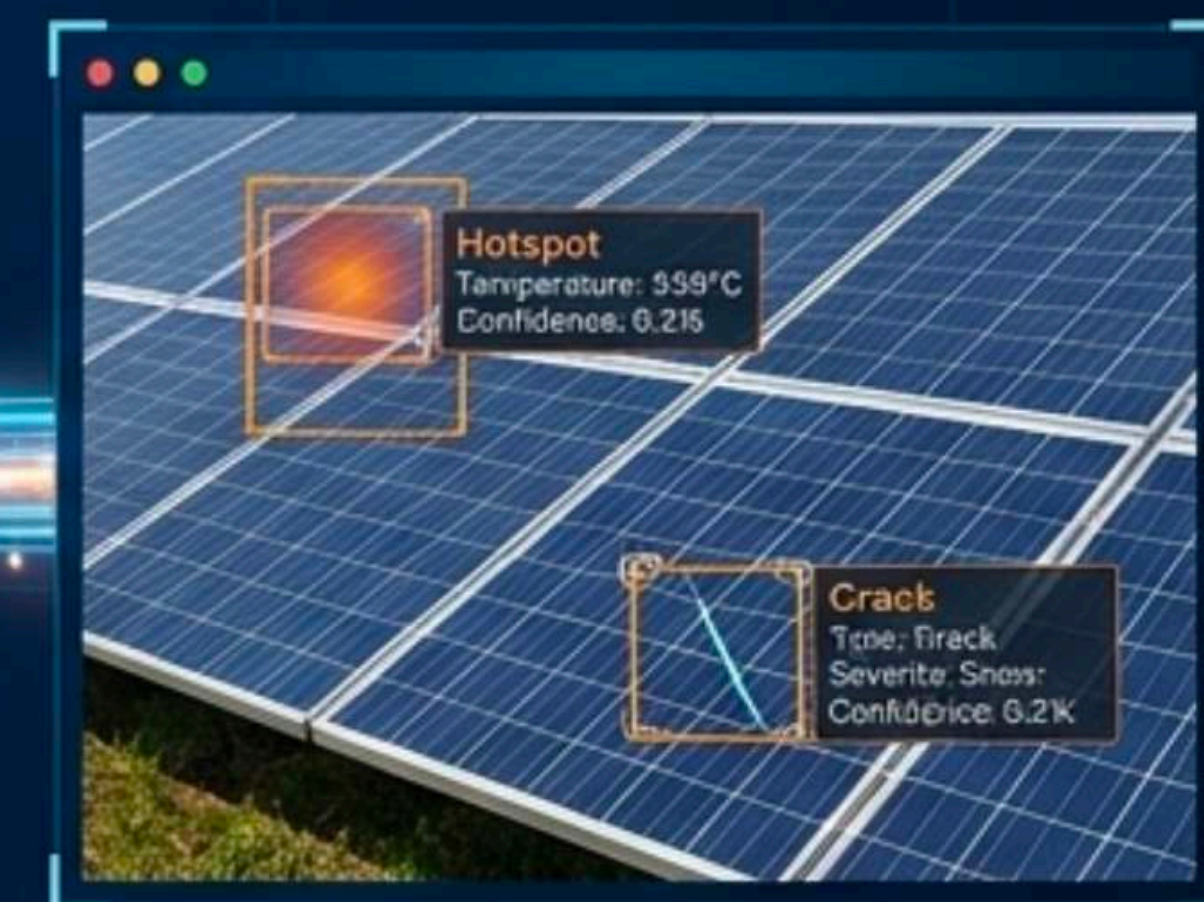
Visible Light Input



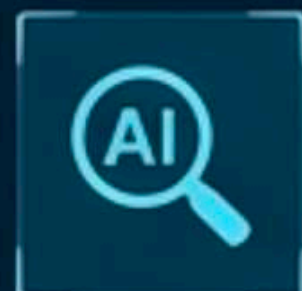
Thermal Input
(Image Fusion)



DarkNet53



AI detection dashboard



Feature Extraction

Neural network recognition of key issues (Hotspots, micro-cracks, dust).



Classification Accuracy

Automated detection of damage type and its severity level.

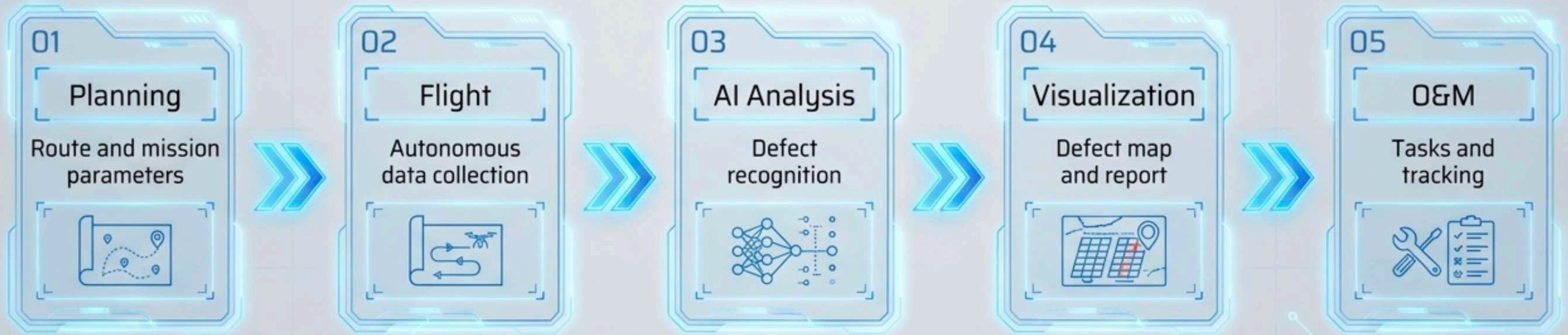


Multi-Scale Fusion

Simultaneous analysis of macro-objects and microscopic defects.

How the System Works: From Flight to Defect Resolution

The system generates an inspection route → the drone performs an autonomous flight → the platform analyzes data → detects defects → displays on a map → generates a report → creates tasks → tracks resolution. Repair results return to the system, becoming part of the asset's history.

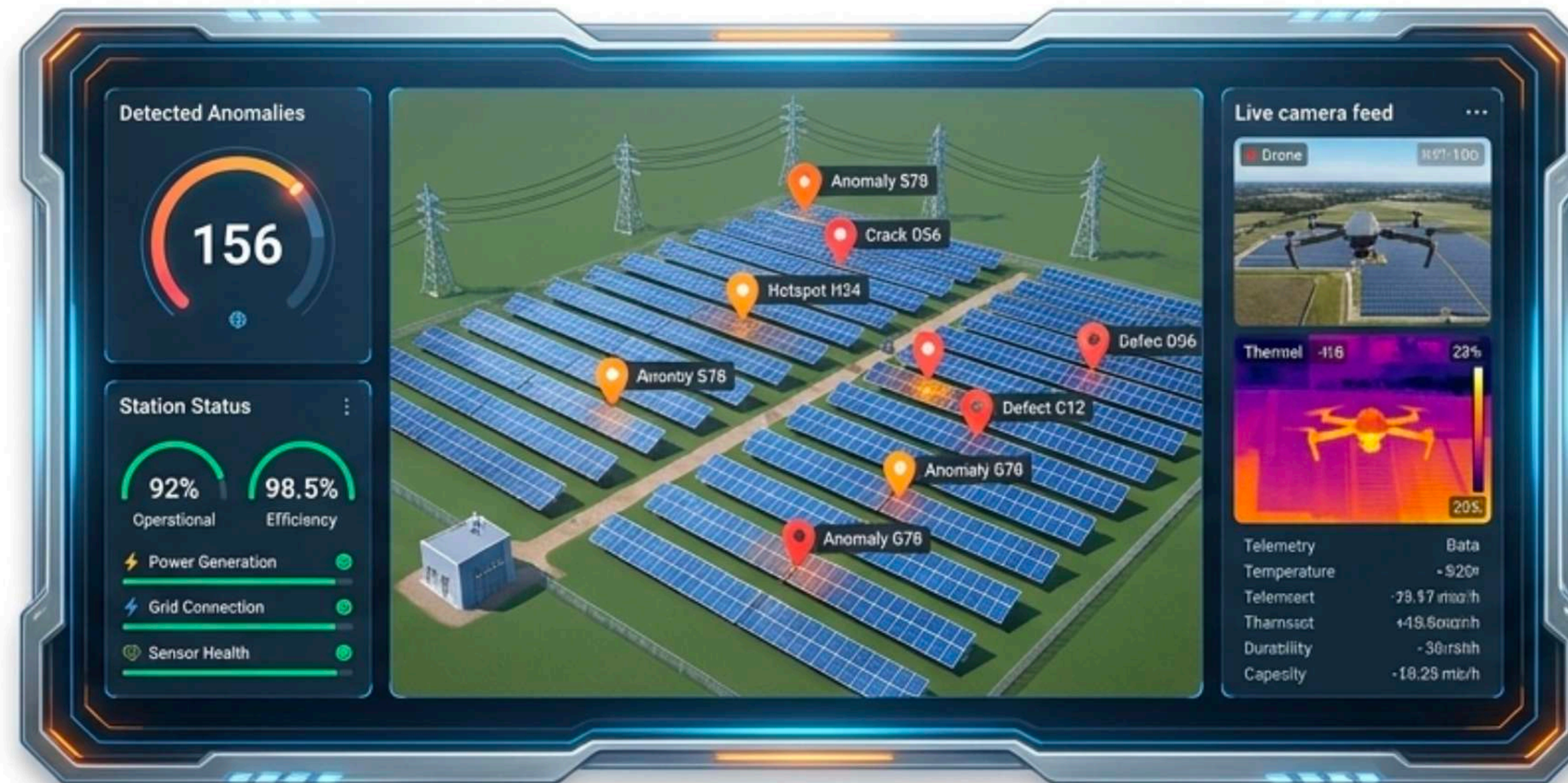


Digital Control Room: Absolute Control



DRONE
SERVICE

UNB
GROUP



2D/3D Visualization

Interactive station model with precise georeferencing for every damaged panel.



Real-time Updates

Dashboard with key performance indicators (KPI) and inspection statuses.



Deep Analytics

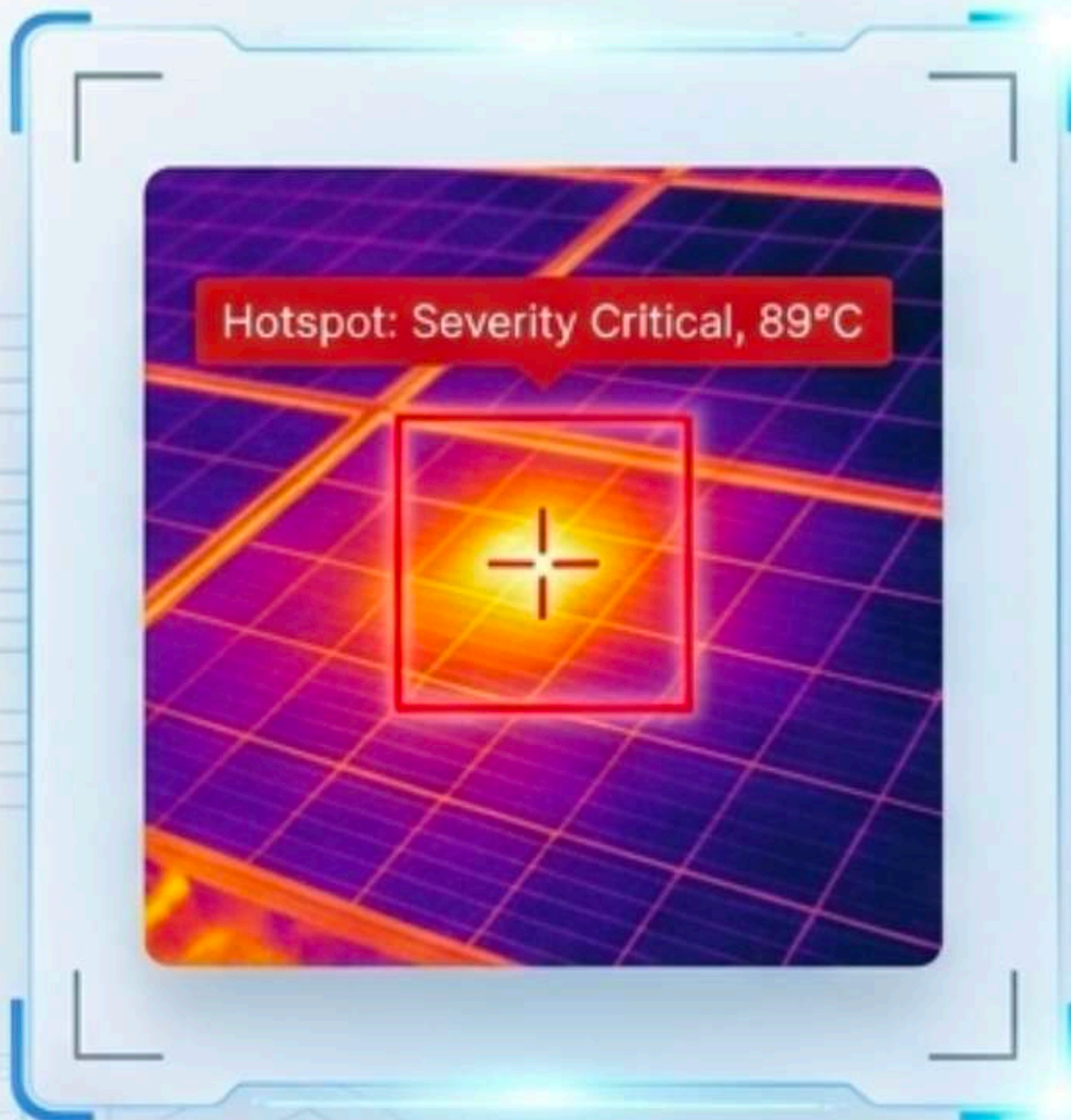
Flight history, period comparison, and asset degradation trends.

From Pixel to Repair: O&M Workflow



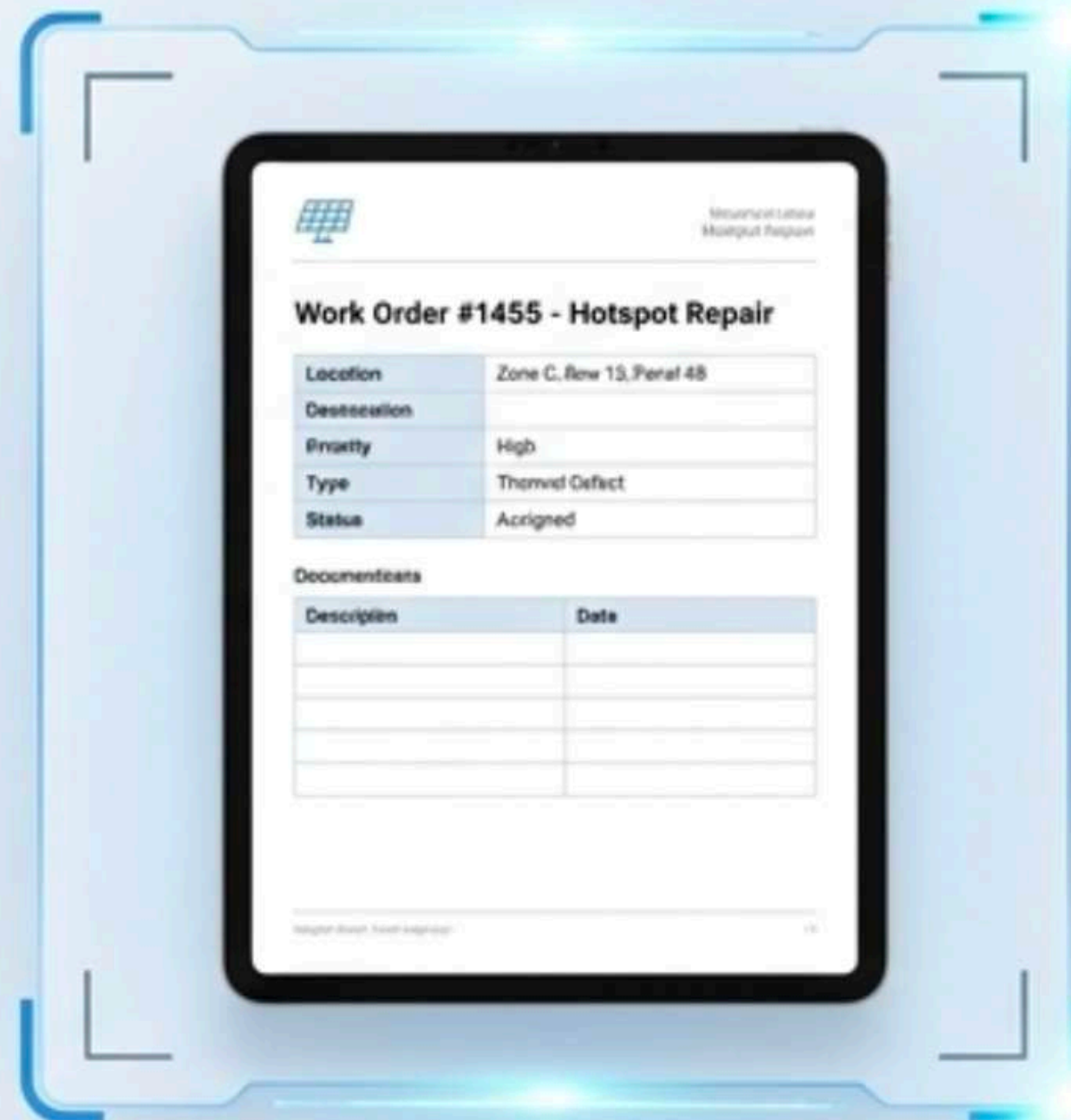
**DRONE
SERVICE**

**UNB
GROUP**



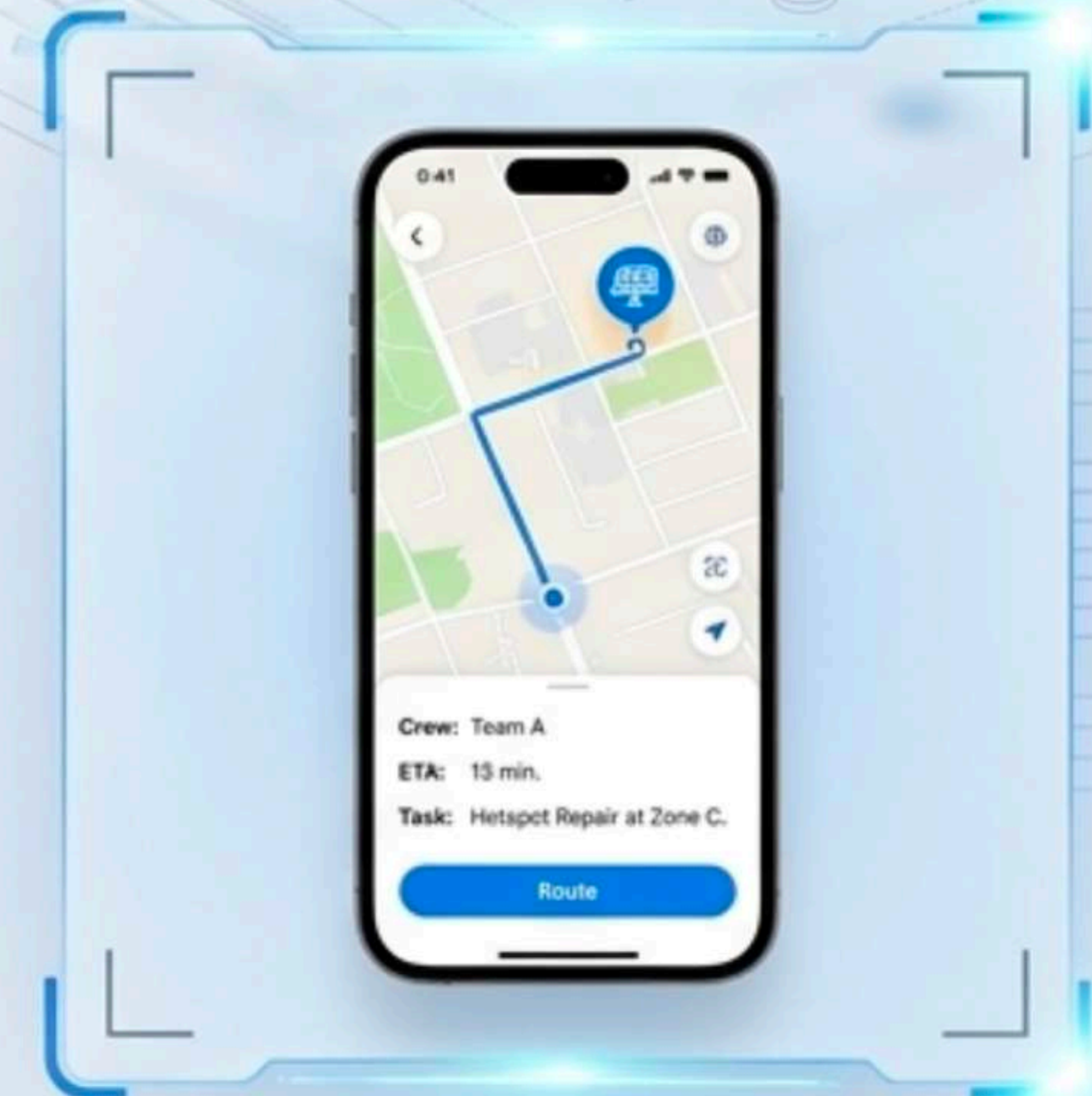
Step 1: Detection

AI identifies the defect and assigns its priority.



Step 2: Task Automation

The platform auto-generates a Work Order with exact GPS coordinates and issue type.



Step 3: Crew Routing

The engineer receives the task in a mobile app with a generated route. Task closure is done via photo report.

Scalability: Managing a Network of Assets



**DRONE
SERVICE**

**UNB
GROUP**

Unified Platform

Connect and control dozens of remote stations from a single point.

Centralized Monitoring

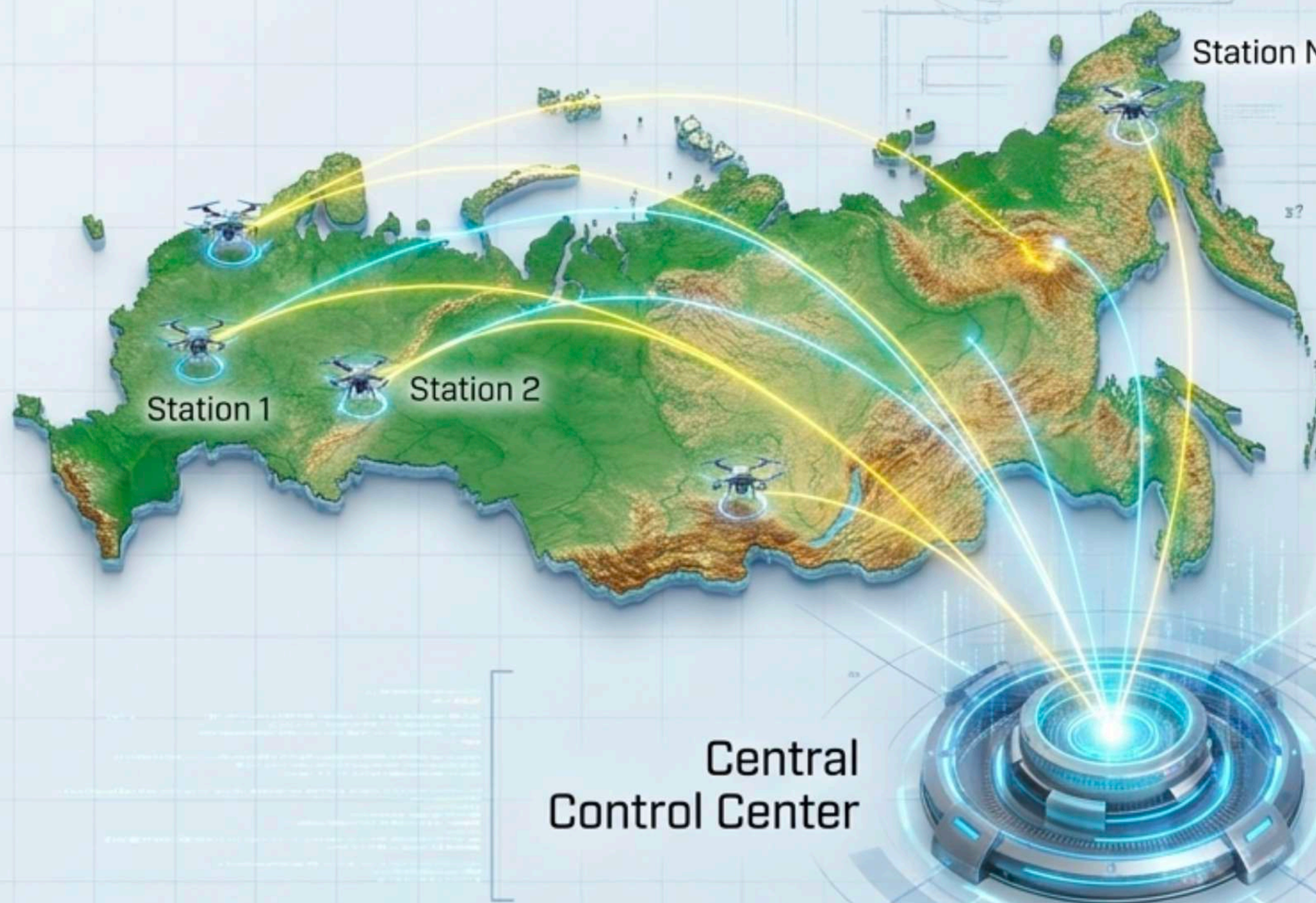
A global dashboard for the entire asset portfolio.

Fleet Management

Synchronized control of multiple UAVs.

Unification

Standardized data, AI analytics, and management reporting across sites.



Defects We Detect

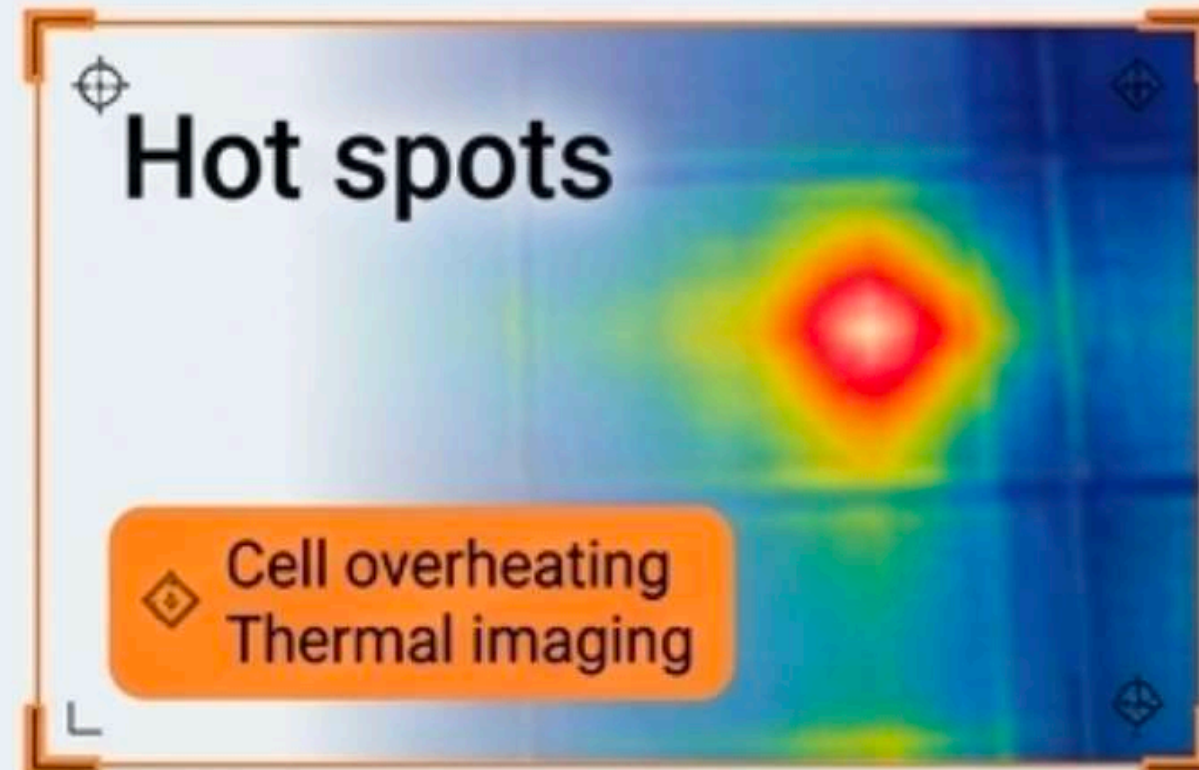
AI-recognition of key solar panel issues



DRONE
SERVICE

UNB
GROUP

AI-recognition of key solar panel issues. The system focuses on practical operational scenarios, automatically identifying the most critical defect types. Analysis is based on visible light, infrared, image fusion, and AI models.



System Performance

11+ Defect Types

95%+ AI Accuracy

<2 min Per Analysis

Impact for the Client

Practical Business Outcomes



**DRONE
SERVICE**

**UNB
GROUP**

The platform makes inspections faster, more accurate, and controllable. By automating flights, early defect detection, and centralized maintenance tracking, operating costs are reduced, generation losses decrease, and O&M transparency increases.



-60%

Inspection Costs

Automation replaces manual labor.



10x

Faster Detection

AI replaces visual inspection.



-15%

Generation Losses

Early defect detection.



100%

O&M Transparency

Complete process visibility.



Less Manual Labor

Autonomous drones and automated data processing reduce staff workload.



Predictive Management

Transition from reactive to proactive, data-driven maintenance.



Scalability

Ready to expand across a network of facilities without a proportional cost increase.

Result: The client receives not just a set of technologies, but a working tool to enhance asset efficiency.

The Future of Solar Energy is Here Today

We offer more than just inspection drones. We offer a unified intelligent platform that combines inspection, analytics, visualization, and maintenance into a single manageable digital process.

UNB group — your reliable partner in asset digitalization.



**DRONE
SERVICE**

**UNB
GROUP**



Contacts

Email: info@unbgroup.uz

Website: unbgroup.uz

Contact Person: Akmal Karimov

Phone: +998 94 388 88 82

UNB
G R O U P



DRONE
SERVICE